

E-Commerce Sales Analytics & Power BI Dashboard

Data Analytics Internship Final Project — a full pipeline from raw e-commerce transactions to an interactive Power BI Sales Performance Dashboard that surfaces actionable insights.

Tools: Python | Pandas | NumPy | Matplotlib | Power BI | DAX | GitHub

Project Overview & Objective

Objective

Analyze e-commerce transaction data to understand sales performance, product/category behavior, monthly trends, order status, payment preferences, and inter-variable relationships.

Final Outcome

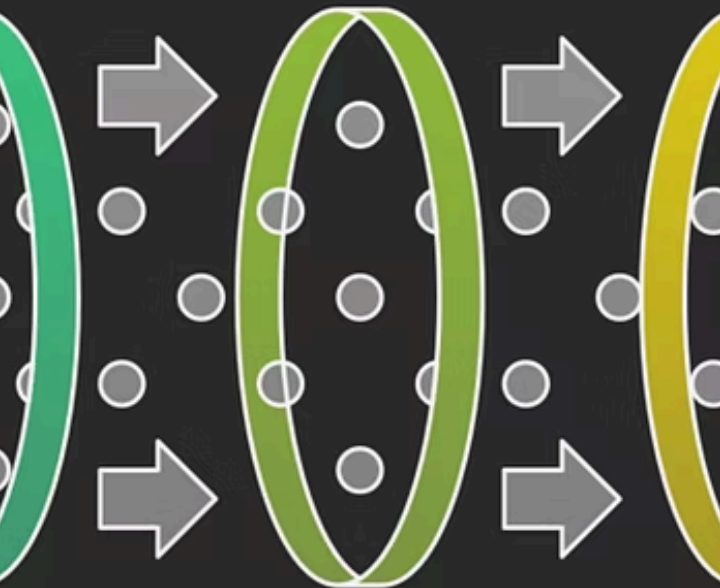
An interactive Power BI Sales Performance Dashboard that converts raw transactions into actionable business insights for stakeholders.

Dataset

The project uses a simulated e-commerce transaction dataset containing Customers, Products, Categories, Orders, Prices, Discounts, Shipping, Payment methods, and Order status — a rich source for transactional and behavioral analysis.

Search					
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tity	Avg_Price	Delivery_Charges	Coupon_Status	GST	Da
1	153.71	6.5	Used	0.1	1/1/2
1	153.71	6.5	Used	0.1	1/1/2
2	122.77	6.5	Not Used	0.1	1/1/2
1	81.5	6.5	Clicked	0.1	1/1/2
1	153.71	6.5	Clicked	0.1	1/1/2
1	153.71	6.5	Clicked	0.1	1/1/2
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1	256.88	6.5	Used	0.1	1/1/2
1	153.71	28.78	Clicked	0.1	1/1/2
1	122.77	6.5	Used	0.1	1/1/2
1	153.71	6.5	Not Used	0.1	1/1/2
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1	153.71	6.5	Used	0.1	1/1/2
1	122.77	6.5	Not Used	0.1	1/1/2

Data Cleaning Process Funnel



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Evaluate Quality
Assess data accuracy and reliability

Act o
Correct data

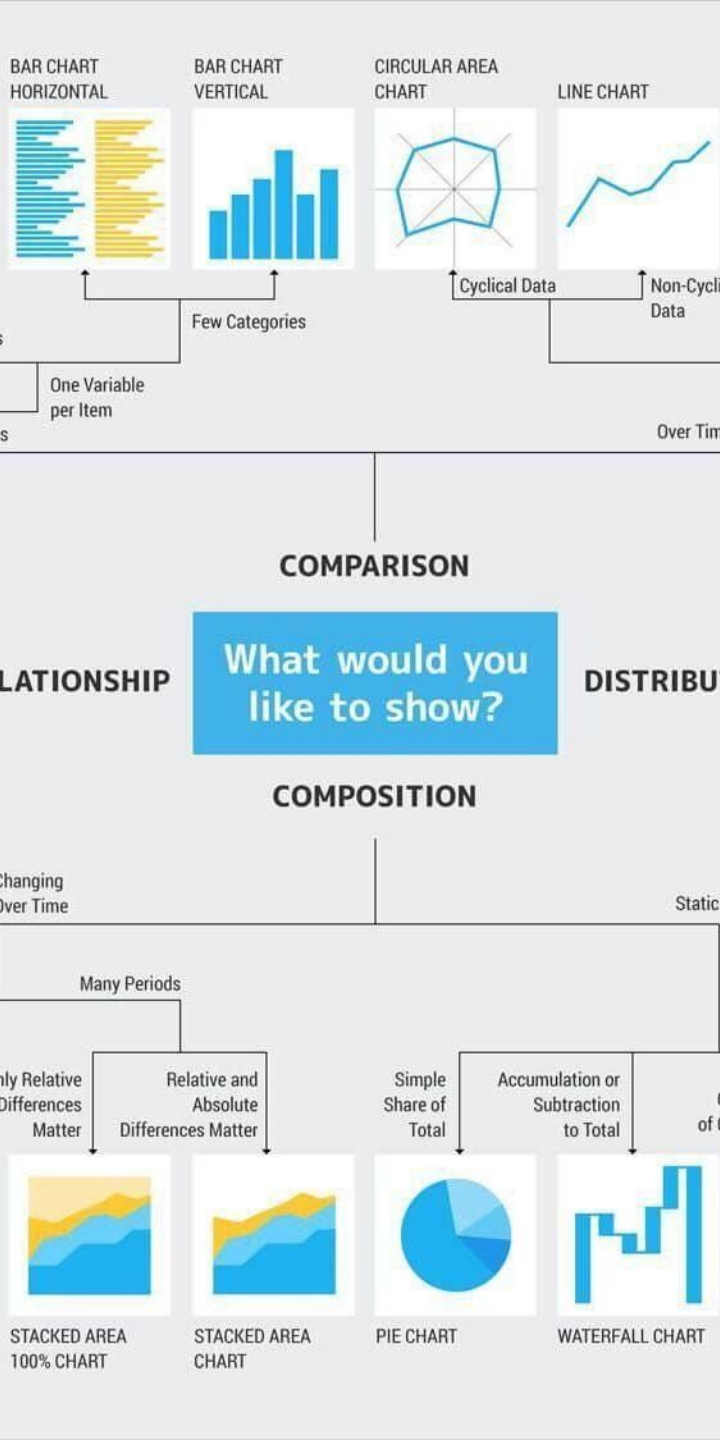
Data Preparation & Feature Engineering

Cleaning Steps

- Identified and handled missing values
- Removed duplicates and invalid quantities/prices
- Converted dates and data types

Engineered Metrics

Created Gross Sales, Discount Amount, and Net Sales to enable accurate revenue and margin analysis.



Exploratory Data Analysis (EDA)



Descriptive & Category Analysis

Summary stats and category-level comparisons to reveal which categories drive revenue.



Monthly Trends

Monthly sales analysis highlighted fluctuations and seasonal patterns.



Order & Payment Analysis

Order-status and payment-method analyses uncovered delivery success rates and preferred payment channels.



Correlation & Outliers

Correlation checks and outlier detection informed data quality and analytical assumptions.

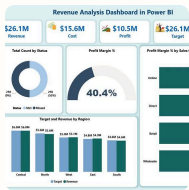


Key Visualizations Built

Major visualizations included Net Sales by Product Category, Monthly Net Sales Trend, Top 10 Products by Net Sales, Order Status Distribution, Orders by Payment Method, and Quantity vs Net Sales — each designed to clarify trends, rankings, and relationships.

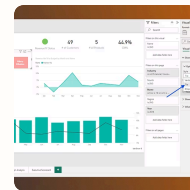
📊 Visuals were chosen to support fast, interactive exploration in Power BI.

Power BI Dashboard Design & Interactivity



KPI Cards

Total Net Sales, Total Orders, Total Quantity Sold, Average Order Value — displayed prominently for at-a-glance performance tracking.



Interactive Filters

Product Category, Order Status, Order Date, and Payment Method slicers enable dynamic cross-filtering and rapid hypothesis testing.

Key Insights & Recommendations

Insights

- Electronics is the top revenue-generating category; high-value Electronics dominate top rankings.
- Monthly sales exhibit noticeable fluctuations, indicating demand variability.
- Most orders are delivered successfully; cancellations and returns need attention.
- UPI and Credit Card are major payment channels.

Recommendations

1. Maintain inventory and run targeted promotions for top Electronics products.
2. Investigate root causes of cancellations and returns; implement remediation.
3. Prioritize UPI and Credit Card promotions to increase conversion and loyalty.
4. Leverage monthly demand patterns for inventory and marketing planning.



or Wo...



Jse ... ?



VALENTINE Local 2026 New 7-inch Wi...

\$37.44 ~~\$92.80~~ 🔥 11K+ sold

TOP RATED in Car Audio & Video ?

★★★★★ 1,453

★ Star store

Arrives in 2+ business days



VALENTINE

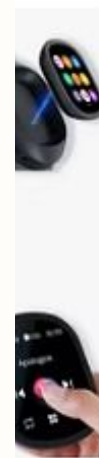
\$29.15 ~~\$~~

TOP RATE

★★★★★

Brand C

Arrives in



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ion... ?



VALENTINE Local Smart Clock Camera,...

\$28.20 ~~\$59.99~~ 🔥 1.3K+ sold

BEST-SELLING ITEM

★★★★★ 192



VALENTINE

Last day

SELL

★★★★★

Made with GAMMA

Conclusion & Next Steps

This project demonstrates a complete Data Analytics → Business Intelligence → Decision-Making workflow. Data-driven analysis identified revenue drivers, operational issues, customer payment preferences, and opportunities for growth.

Technologies: Python + Power BI + DAX + Git/GitHub

Thank you — Questions?